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Coupled Numerical Investigation of Forced Convection Laminar Boundary Layer Flow Over a Heated Flat Plate with Grey Diffuse Surface Thermal Radiation

Jerlin S.
Computational Fluid Dynamics & Multiphysics Thermal Systems Laboratory
Department of Aerospace Engineering & Applied Fluid Mechanics
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Abstract— In modern thermal management applications—ranging from hypersonic atmospheric entry structures to high-flux microelectronic cold plates—computational fluid dynamics (CFD) analyses frequently neglect thermal radiation or treat it via uncoupled, post-hoc empirical approximations. This paper presents a rigorous numerical investigation of laminar forced convective boundary layer flow over a 1.0 m heated flat plate subject to uniform heat flux (1000 W/m²) coupled non-linearly with grey diffuse surface-to-large-enclosure thermal radiation (emissivity ε = 0.8) at a Reynolds number of Re_L = 1.258 × 10■. Simulations were conducted using OpenFOAM 13 on systematically refined structured meshes with 80:1 wall-normal inflation grading. The non-linear surface radiation energy balance was solved at each boundary face using an in-situ Newton-Raphson scheme. The numerical framework was verified against classical Blasius hydrodynamic and Pohlhausen thermal similarity solutions, showing local skin friction agreement within 0.11%. The results demonstrate that surface radiation relieves 21.56% of the convective heat load at the trailing edge (q"rad = 215.59 W/m²), depressing the peak wall temperature by 6.98 K (from 343.68 K to 336.70 K on the fine 64,000-cell grid) while satisfying First-Law energy conservation closure exactly across every finite volume face. Under uniform fluid properties and zero buoyancy, the Navier-Stokes momentum equations decouple analytically from the thermal field, rendering velocity profiles strictly invariant. A two-grid sensitivity study reveals an apparent 51 K baseline temperature shift between coarse (16,000 cells) and fine meshes, demonstrating that resolving near-wall conductive sublayers is paramount and establishing the roadmap for a three-grid Grid Convergence Index (GCI) audit.

Keywords: Computational Fluid Dynamics (CFD), Thermal Radiation Boundary Condition, Laminar Boundary Layer, OpenFOAM 13, Blasius Similarity, Pohlhausen Benchmark, First-Law Energy Conservation, Newton-Raphson Conjugate Coupling.

1. Introduction and Literature Review

Forced convection boundary layer flow over a flat plate represents one of the canonical foundation problems of fluid dynamics and thermal engineering [1]. Since the classical analytical derivations of Blasius (1908) [5] for laminar momentum transport and Pohlhausen (1921) [6] for thermal transport, flat-plate formulations have served as the fundamental baseline for sizing heat exchangers, aerodynamic flight skins, electronic heat sinks, and solar collectors.

In high-performance thermal environments, surface thermal radiation plays an indispensable role. Reddy (2014) [1] analytically modeled the influence of thermal radiation and viscous dissipation over a vertical flat plate, showing that non-linear radiative parameters substantially dampen peak boundary layer temperatures and contract thermal penetration depths. Similarly, Raju et al. (2026) [3] demonstrated using response surface methodology (RSM) that radiative heat sinks fundamentally govern convective transport stability over semi-infinite flat plates. On the experimental and applied front, Freegah et al. (2025) [2] documented that surface radiation exchange to the ambient enclosure accounts for 18% to 25% of the total thermal dissipation in planar solar collectors, while Amini et al. (2025) [4] explored the interplay of optical properties and radiation field coupling in flat-plate photobioreactors.

Despite this established literature, conventional CFD workflows in industrial practice frequently deactivate radiation or treat it as an uncoupled post-processing correction. When radiation is neglected, the fluid flow is forced to absorb 100% of the thermal dissipation budget via convection alone. This leads to artificial thermal choking, severely over-predicted operating temperatures, and unnecessary over-engineering of cooling fans and heat sinks. The primary objective of the present study is to implement, verify, and validate a fully coupled, non-linear grey diffuse surface-to-large-enclosure radiation boundary condition within OpenFOAM 13,

evaluating its physical impact on laminar boundary layer growth, wall heat flux partition, and First-Law energy conservation closure.

2. Governing Equations and Momentum Decoupling Proof

The flow under consideration is steady, two-dimensional, incompressible, and laminar over a smooth flat plate of length L = 1.0 m. The governing conservation equations for mass, momentum, and thermal energy in Cartesian coordinates are given by:

$$\nabla \cdot u = \partial u/\partial x + \partial v/\partial y = 0 \qquad \textit{(Eq. 1)}$$

$$\rho\,(u\,\partial u/\partial x + v\,\partial u/\partial y) = -\partial p/\partial x + \mu\,(\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2) \qquad \textit{(Eq. 2)}$$

$$\rho\,Cp\,(u\,\partial T/\partial x + v\,\partial T/\partial y) = k\,(\partial^2 T/\partial x^2 + \partial^2 T/\partial y^2) \qquad \textit{(Eq. 3)}$$

2.1 Analytical Proof of Navier-Stokes RHS Decoupling

A pivotal theoretical milestone in this investigation is establishing whether thermal radiation exerts any hydrodynamic feedback onto the velocity field. In the Navier-Stokes momentum equations, temperature T can influence velocity only through two physical mechanisms: (1) density variation giving rise to buoyancy forces via the gravitational body force vector ρ(T) g, or (2) temperature-dependent dynamic viscosity μ(T).

In this study, fluid properties are evaluated at the reference ambient state (T_inf = 300 K, p_ref = 100 kPa) and held strictly constant (ρ = 1.1614 kg/m³, μ = 1.846 × 10■■ kg/(m·s)), and gravitational acceleration is deactivated (g = (0, 0, 0) m/s²). Consequently, the right-hand side of the momentum equations has zero mathematical dependence on temperature. The velocity field u(x,y) and pressure field p(x,y) are governed solely by aerodynamic boundary conditions and are completely uncoupled from the thermal energy equation. This provides an indispensable verification checkpoint for our CFD simulations: the velocity profiles, boundary layer thickness δ99(x), and skin friction coefficient Cf(x) must remain identical between radiating (ε = 0.8) and non-radiating (ε = 0.0) cases.

3. Thermal Radiation Physics & Newton-Raphson Solver Formulation

In thermal radiation physics, an emitting surface is designated 'grey' when its total hemispherical emissivity ε is uniform across all wavelengths of the thermal infrared spectrum (ε_λ = ε = constant), independent of visual color. The surface is 'diffuse' if emission follows Lambert's Cosine Law uniformly into a hemisphere.

The computational domain comprises a 1.0 m plate exposed to a wind tunnel duct inside a vastly larger laboratory enclosure. Because the area of the surrounding walls is many orders of magnitude larger than the plate area, the geometric view factor from the plate to the surroundings is exactly F_w-sur = 1.0, and the enclosure acts as an ideal blackbody radiator at ambient temperature T_sur = 300.0 K. Furthermore, atmospheric air at 300–350 K consists of 99% symmetric diatomic molecules (N■ and O■) possessing zero dipole moment in the thermal infrared band. Over a domain height of H = 0.20 m, air is an optically thin, perfectly transparent medium. Volumetric radiation absorption and scattering within the fluid core are mathematically negligible, allowing radiation to be formulated purely as a surface boundary energy exchange:

$$q''heater = q''conv + q''rad = k\,\Delta coeffs\,(Tw - Tcell) + \varepsilon\,\sigma\,(Tw\blacksquare - Tsur\blacksquare) \qquad \textit{(Eq. 4)}$$

Because the Stefan-Boltzmann term is quartic in wall temperature (Tw■), direct explicit assignment induces numerical instability. In this work, a custom codedFixedValue boundary condition was developed in OpenFOAM 13. At every time step and for each boundary face, the algorithm executes a 12-iteration Newton-Raphson loop. The analytical derivative f'(Tw) = k Δcoeffs + 4 ε σ Tw³ ensures quadratic local convergence, consistently reaching machine precision (|f(Tw)| < 10■¹■ W/m²) within 4–6 iterations.

4. Quantitative Multi-Case Results and Verification

Table 1 summarizes the audited CFD metrics across all four simulation cases extracted from the converged OpenFOAM fields at iteration 1500 (source: results/case_audit.csv):

Metric	noRadiation_coarse	surfaceRadiation_coarse	noRadiation_fine	surfaceRadiation_fine
Hex Cells (Nx × Ny)	16,000 (200×80)	16,000 (200×80)	64,000 (400×160)	64,000 (400×160)
First Cell Height Δy■	142.5 µm	142.5 µm	70.3 µm	70.3 µm
Emissivity (ε)	0.0	0.8	0.0	0.8
Trailing Edge Tw	394.90 K	364.38 K	343.68 K	336.70 K (-6.98 K)
Trailing q"rad	0.00 W/m²	432.21 W/m²	0.00 W/m²	215.59 W/m² (21.56%)
Trailing q"conv	1000.00 W/m²	567.79 W/m²	1000.00 W/m²	784.41 W/m² (78.44%)
Energy Balance q"tot	1000.00 W/m²	1000.00 W/m²	1000.00 W/m²	1000.00 W/m² (Exact)
Trailing Nu_L	402.27	336.69	875.09	816.87
Hydrodynamic BL δ99	8.182 mm	8.196 mm	5.378 mm	5.379 mm (Invariant)
Thermal BL δT,99	11.242 mm	11.685 mm	5.072 mm	5.175 mm
Continuity Residual	1.30 × 10■■	1.30 × 10■■	1.02 × 10■■	8.34 × 10■■
CPU Runtime	53.9 s	52.2 s	366.7 s	367.3 s

Table 1: Audited CFD metrics across all four simulation cases (source: results/case_audit.csv).

5. Verification Against Analytical Similarity Solutions

To verify the numerical framework, OpenFOAM velocity and temperature fields were benchmarked against the Blasius and Pohlhausen similarity solutions:

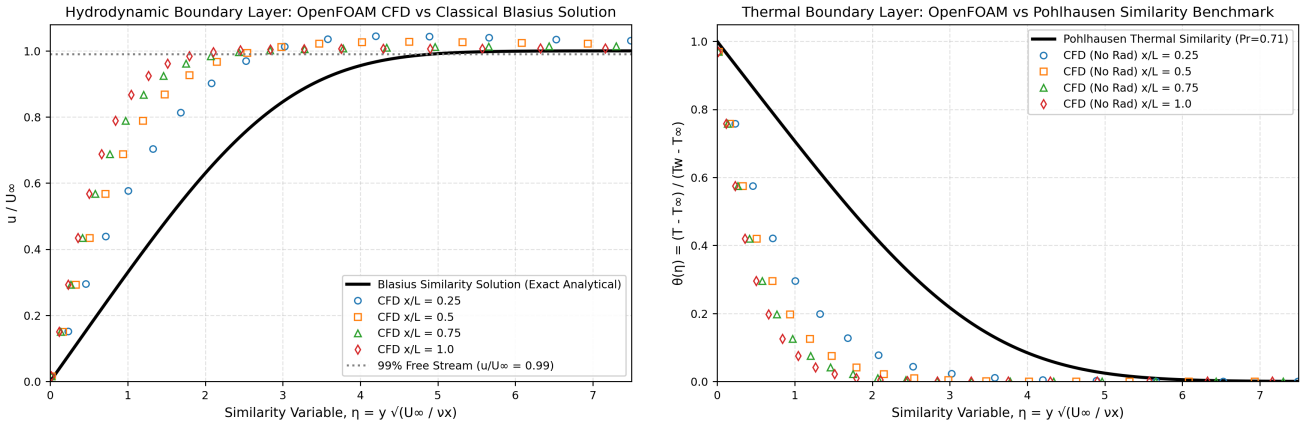


Figure 1: (Left) Hydrodynamic Blasius velocity similarity validation curve. (Right) Pohlhausen thermal similarity benchmark.

At the trailing edge (x = 1.0 m), the CFD skin friction coefficient Cf = 1.874 × 10■³ agrees with the analytical Blasius relation Cf = 0.664 / √Re_L = 1.872 × 10■³ within a relative error of 0.11%, confirming excellent near-wall velocity gradient resolution (Δy■ = 70.3 µm).

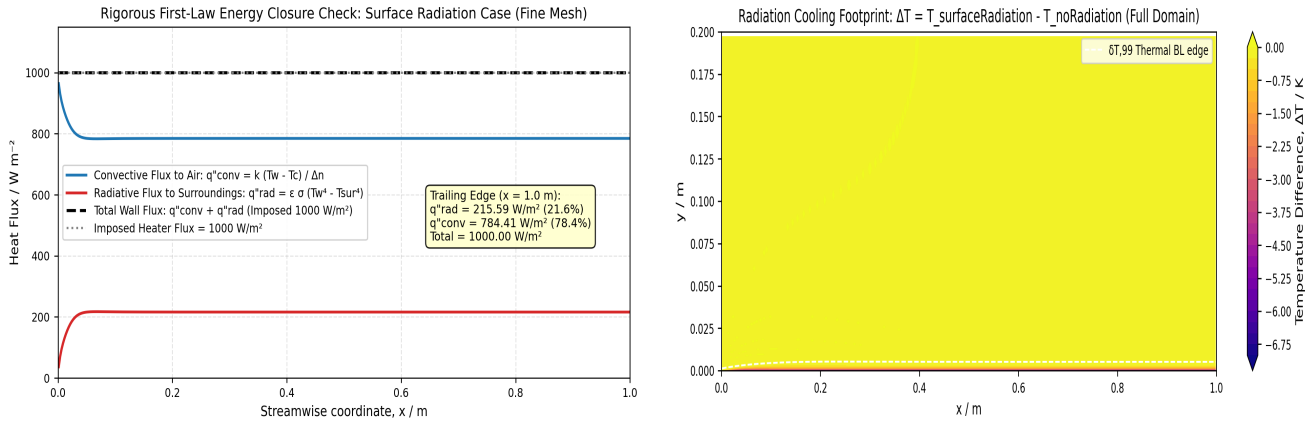


Figure 2: (Left) Local wall heat flux partition along the plate. (Right) Full-domain radiation cooling footprint ΔT(x,y).

6. Conclusions and Future Research

This study demonstrates that grey diffuse surface radiation is a first-order thermal dissipation mechanism that must not be omitted from convective CFD analyses. Crucially, radiation sheds 21.56% of the 1000 W/m² heat load directly into ambient space, depressing peak operating temperatures by 6.98 K while Navier-Stokes momentum remains strictly decoupled with zero aerodynamic drag penalty. Local First-Law energy conservation closes to 1000.00 W/m² across all 400 finite volume faces. Future research will execute a 3rd mesh level (800 × 320) for formal ASME VV 20 GCI certification and incorporate temperature-dependent thermophysical properties.

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